Control Theory

Chap 4: PID Tuning and Modification

Prof. *Guang Zhai*, Assistant. Prof. *Yiyong Sun*

gzhai@bit.edu.cn, sunyy@bit.edu.cn
Rm 403, Bld. Aerospace

November 10, 2026



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

Chap 4: PID Tuning and Modification

1. Introduction
2. Ziegler - Nichols Rules
3. Design of PID Controllers with Frequency - Response Approach
4. PID Controllers with Computational Optimization Approach
5. Modification of PID Control Schemes
6. Conclusion

北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

1 Introduction

2 Ziegler - Nichols Rules

3 Design of PID Controllers with Frequency - Response Approach

4 PID Controllers with Computational Optimization Approach

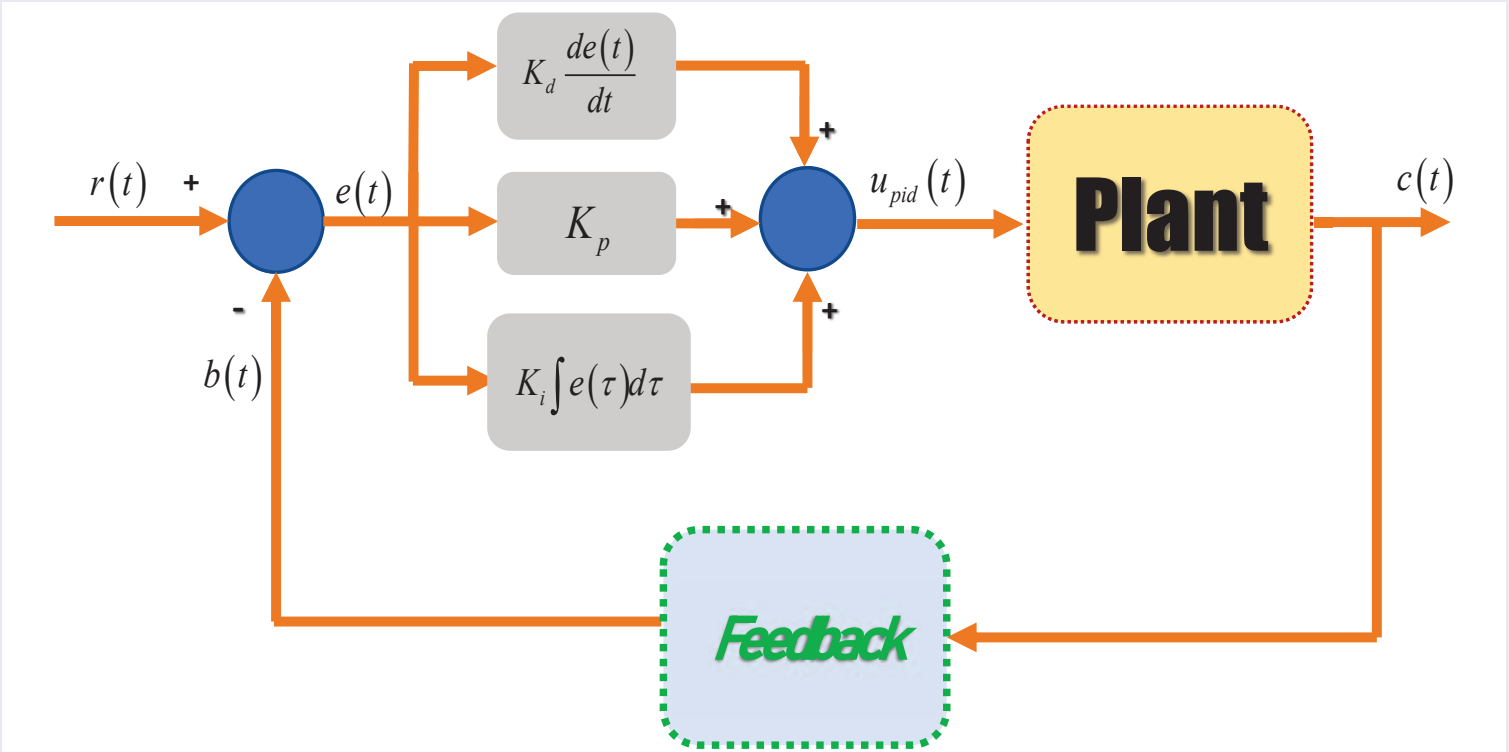
5 Modification of PID Control Schemes

6 Conclusion

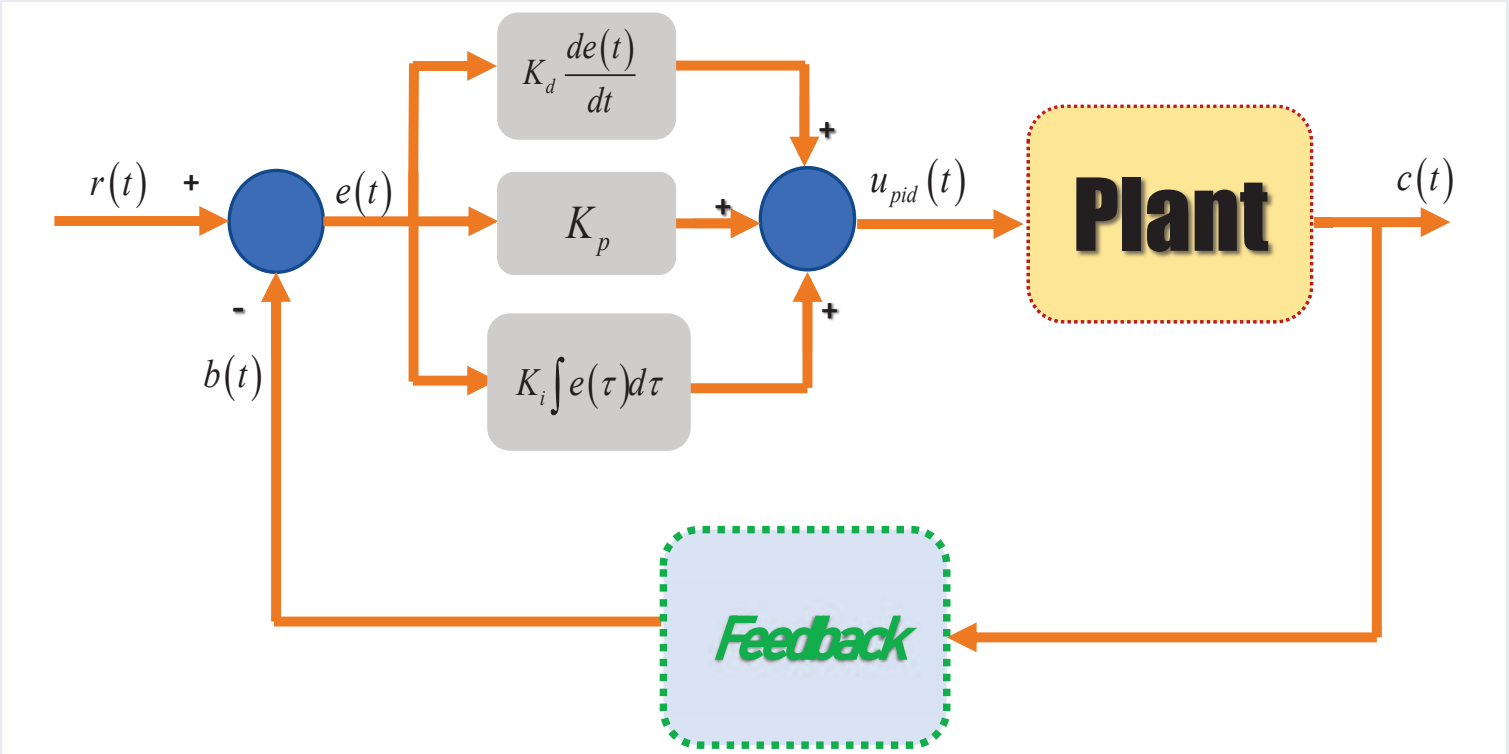


北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

What's PID


$K_d \frac{de(t)}{dt}$
$r(t)$
+
$e(t)$
K_p
+
+
$u_{pid}(t)$
Plant
$c(t)$
+
-
$b(t)$
$K_i \int e(\tau) d\tau$
+
Feedback

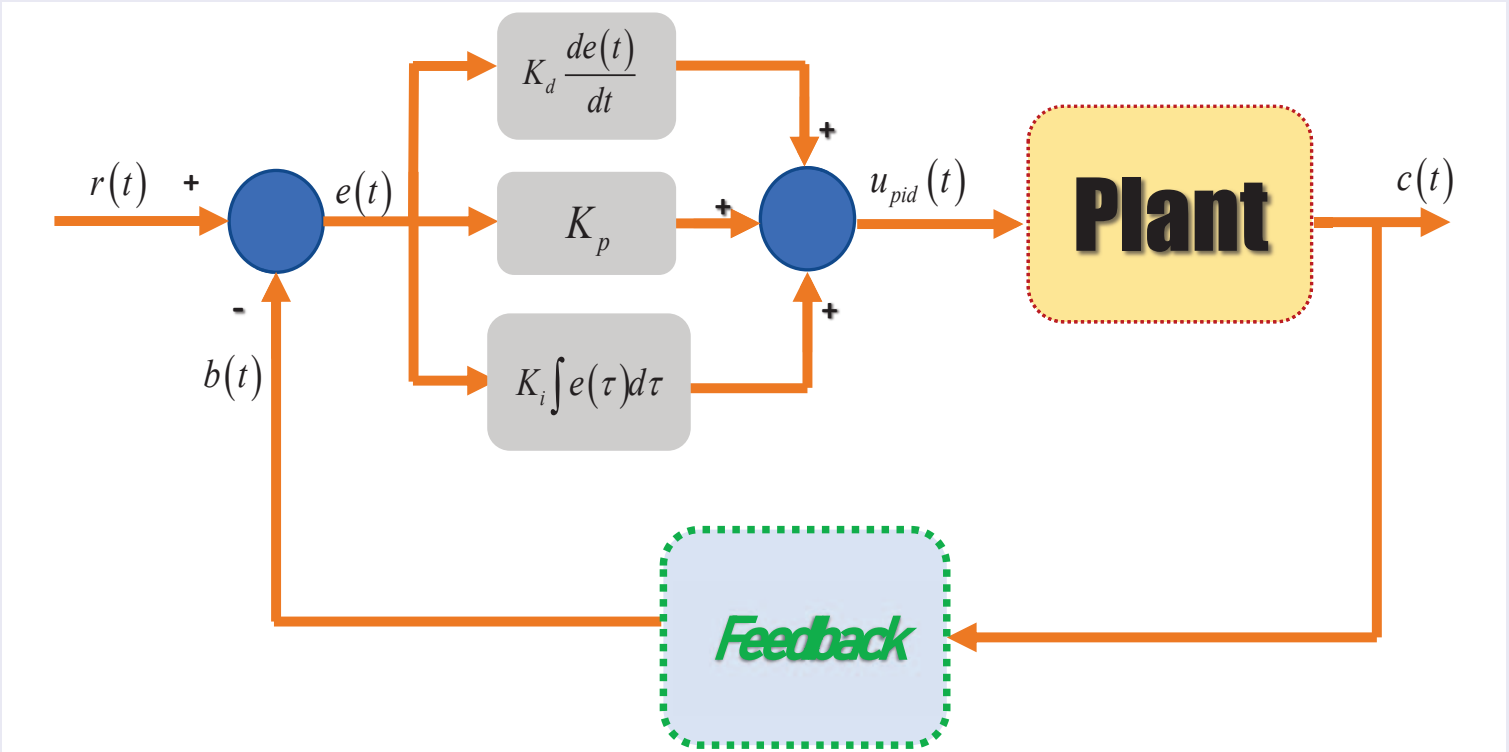
What's PID



Proportional+Integral+Derivative -PID

$$u_{pid}(t) = K_p e(t) + K_i \int e(\tau) d\tau + K_d \frac{de(t)}{dt} \tag{1}$$

What's PID



Proportional+Integral+Derivative -PID （P，PD or PI）

$$u_{pid}(t) = K_p e(t) + K_i \int e(\tau)d\tau + K_d \frac{de(t)}{dt} \tag{1}$$

Laplace Formation of PID

$$\frac{U_{pid}(s)}{E(s)} = K_p + K_i \frac{1}{s} + K_d s \tag{2}$$

Laplace Formation of PID

$$\frac{U_{pid}(s)}{E(s)} = K_p + K_i\frac{1}{s} + K_d s \tag{2}$$

Why PID

- Wide Application: >50% controllers;
- Basis for the other intelligent controllers;
- Can be used for the Model-based and Model-free systems.

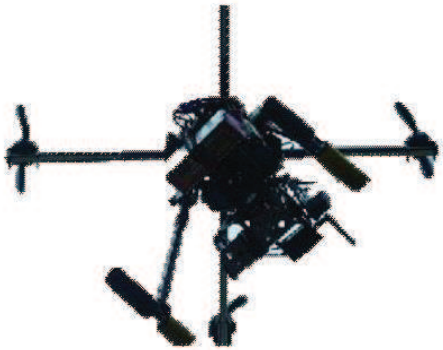
Laplace Formation of PID

$$\frac{U_{pid}(s)}{E(s)} = K_p + K_i\frac{1}{s} + K_d s \tag{2}$$

Why PID

- Wide Application: >50% controllers;
- Basis for the other intelligent controllers;
- Can be used for the Model-based and Model-free systems.







How to design or select the gains?



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

How to design or select the gains?

K_p, K_i and K_d



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

1 Introduction

2 Ziegler - Nichols Rules

3 Design of PID Controllers with Frequency - Response Approach

4 PID Controllers with Computational Optimization Approach

5 Modification of PID Control Schemes

6 Conclusion



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

Instead of using the **PID** controller in (1) or (2), Ziegler-Nichols rules require to have a reformation

$$\frac{U_{pid}(s)}{E(s)} = K_p \left(1 + \frac{1}{T_i s} + T_d s\right) \tag{3}$$

Instead of using the **PID** controller in (1) or (2), Ziegler-Nichols rules require to have a reformation

$$\frac{U_{pid}(s)}{E(s)} = K_p \left(1 + \frac{1}{T_i s} + T_d s\right) \tag{3}$$

What's the motivation for the reformation?

Instead of using the **PID** controller in (1) or (2), Ziegler-Nichols rules require to have a reformation

$$\frac{U_{pid}(s)}{E(s)} = K_p\left(1 + \frac{1}{T_i s} + T_d s\right) \tag{3}$$

What's the motivation for the reformation?

Remark

The ZN rules are not appropriate for all kinds of system. Especially for the high order system.

Instead of using the **PID** controller in (1) or (2), Ziegler-Nichols rules require to have a reformation

$$\frac{U_{pid}(s)}{E(s)} = K_p\left(1 + \frac{1}{T_i s} + T_d s\right) \tag{3}$$

What's the motivation for the reformation?

Remark

The ZN rules are not appropriate for all kinds of system. Especially for the high order system.

2 Typical Ziegler-Nichols Methods

- 1st ZN method: 'S' shape curve, no integrator and dominant complex-conjugate poles under unit-step input.
- 2nd ZN mehtod: The system response can keep oscillations or nearly oscillations just by changing K_p.



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

1 Introduction

2 Ziegler - Nichols Rules
- Ziegler - Nichols First Method
- Ziegler - Nichols Second Method

3 Design of PID Controllers with Frequency - Response Approach

4 PID Controllers with Computational Optimization Approach

5 Modification of PID Control Schemes

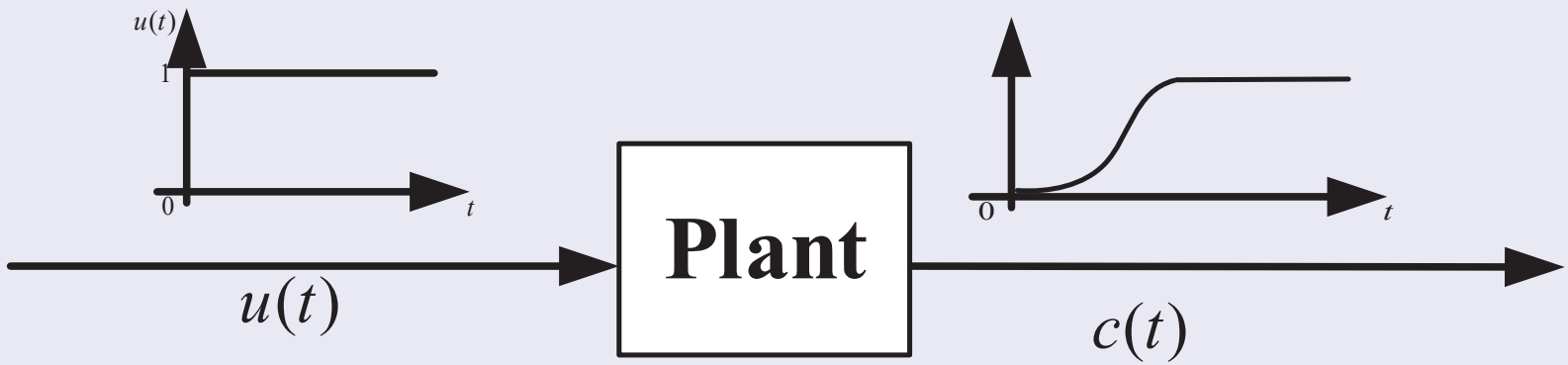
6 Conclusion



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

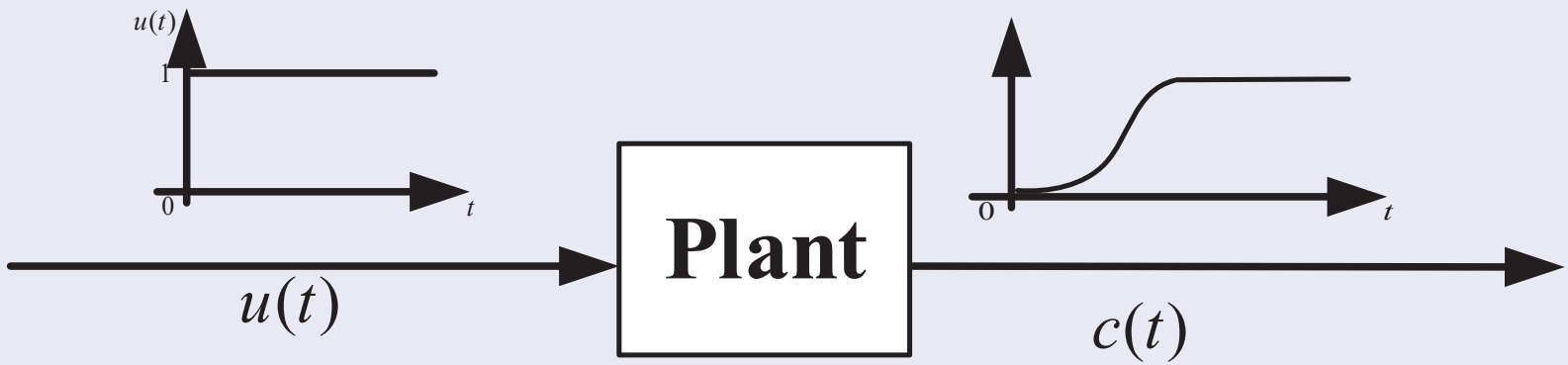
Steps for the First ZN Method

- Step 1: Applying a step input to the plant, determine whether it is in 'S' shape.



Steps for the First ZN Method

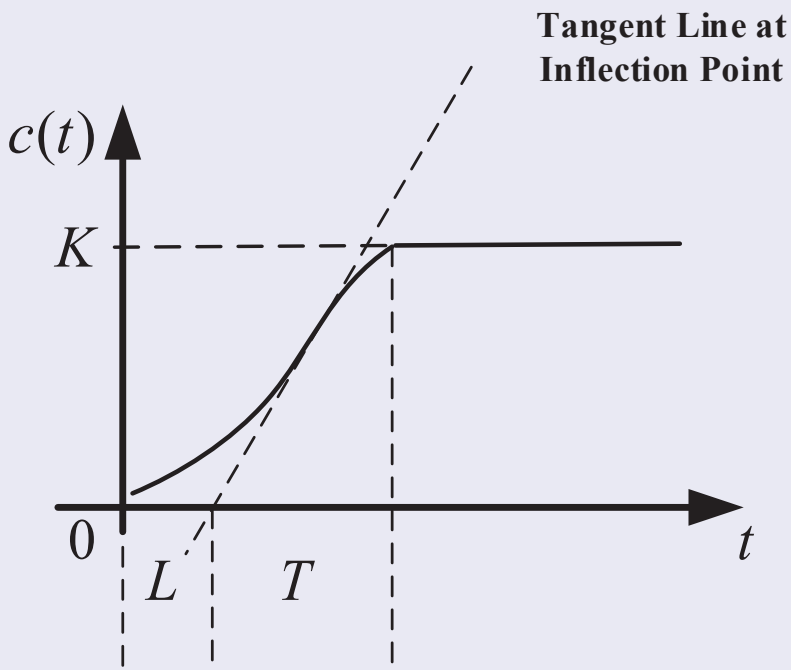
- Step 1: Applying a step input to the plant, determine whether it is in 'S' shape.



If not in 'S' shape, terminate the process.

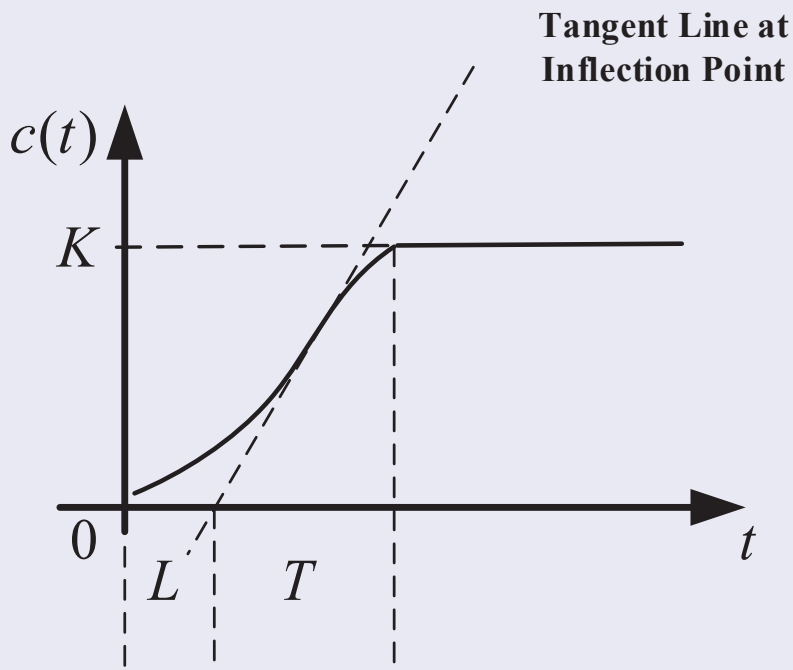
Steps for the First ZN Method

- Step 2: Determine delay time L and time constant T by drawing a tangent line at the inflection point .



Steps for the First ZN Method

- Step 2: Determine delay time L and time constant T by drawing a tangent line at the inflection point .



Such step-response curves may be generated experimentally or from a dynamic simulation.

Steps for the First ZN Method

- Step 3: Obtain the gains K_p, K_i and K_d for P, PI and PID controller

Table 1: Ziegler − Nichols Tuning Rule Based on Step Response of Plant (First Method).

Type of Controller	K_p	T_i	T_d
P	$\frac{T}{L}$	∞	0
PI	$0.9\frac{T}{L}$	$\frac{L}{0.3}$	0
PID	$1.2\frac{T}{L}$	$2L$	0.5L



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

Steps for the First ZN Method

- Step 3: Obtain the gains K_p, K_i and K_d for P, PI and PID controller

Table 1: Ziegler – Nichols Tuning Rule Based on Step Response of Plant (First Method).

Type of Controller	K_p	T_i	T_d
P	$\frac{T}{L}$	∞	0
PI	$0.9\frac{T}{L}$	$\frac{L}{0.3}$	0
PID	$1.2\frac{T}{L}$	$2L$	0.5L

For the PID controller as in 3rd column, it has that

$$\begin{aligned} G_c(s) &= K_p\left(1+\frac{1}{T_i s}+T_d s\right) = 1.2\frac{T}{L}\left(1+\frac{1}{2Ls}+0.5Ls\right) \\ &= 0.6T\frac{\left(s+\frac{1}{L}\right)^2}{s} \end{aligned} \tag{4}$$



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

1 Introduction

2 Ziegler - Nichols Rules
- Ziegler - Nichols First Method
- Ziegler - Nichols Second Method

3 Design of PID Controllers with Frequency - Response Approach

4 PID Controllers with Computational Optimization Approach

5 Modification of PID Control Schemes

6 Conclusion



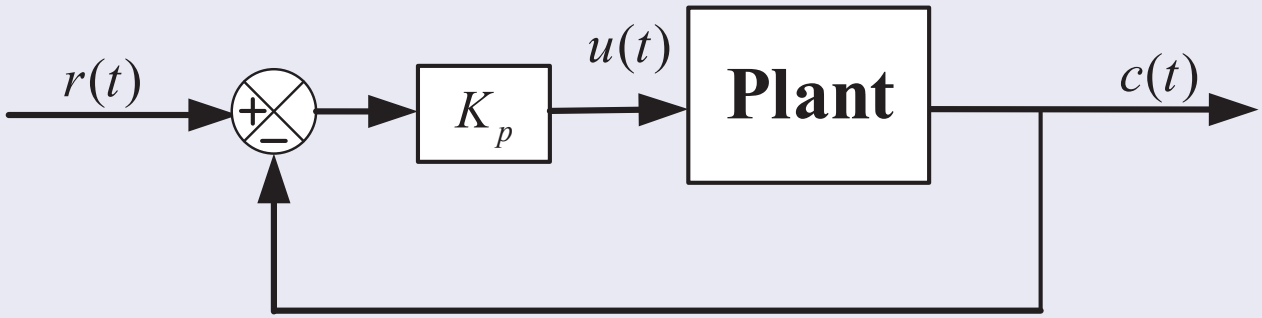
北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

$$\frac{U_{pid}(s)}{E(s)} = K_p\left(1 + \frac{1}{T_i s} + T_d s\right)$$

Z-N Second Method

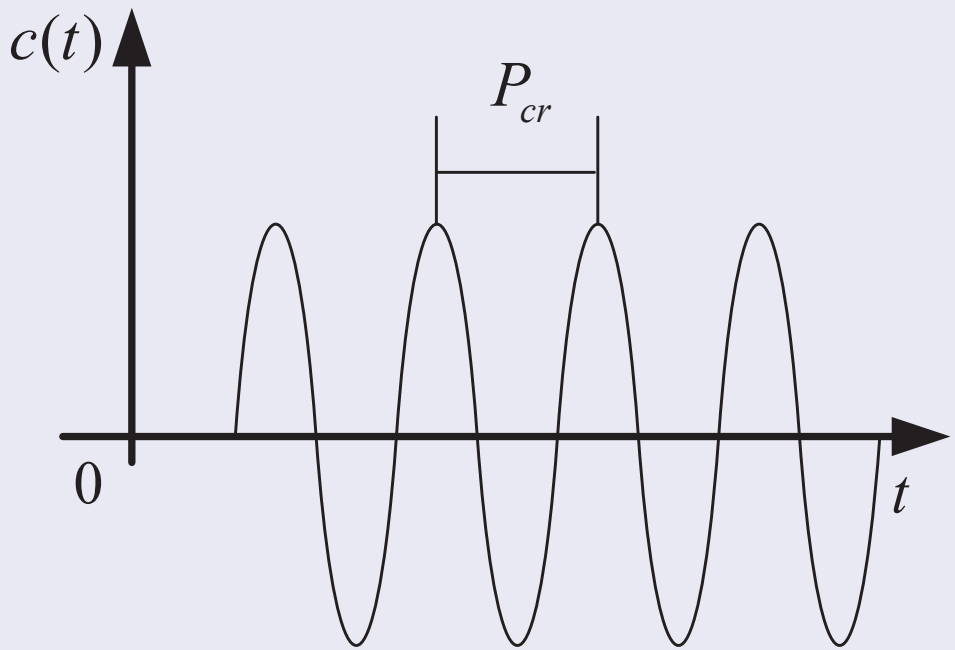
- Step 1: Invalidate the Integral and Differential blocks by setting T_i and T_d to be ∞ and 0.

$$T_i = \infty, T_d = 0, (K_i = 0, K_d = 0)$$



Z-N Second Method

- Step 2: Change the gain K_p , stop until the system exhibits sustained oscillations or nearly approaches oscillations.



Z-N Second Method

- Step 3: With the critical values K_{cr} and P_{cr} validate the sustainable oscillation, one can obtain the PID controller according the Table given below

Table 2: Ziegler – Nichols Tuning Rule Based on Critical Values K_{cr} and P_{cr} (Second Method).

Type of Controller	K_p	T_i	T_d
P	$0.5K_{cr}$	∞	0
PI	$0.45K_{cr}$	$\frac{P_{cr}}{1.2}$	0
PID	$0.6K_{cr}$	$0.5P_{cr}$	$0.125P_{cr}$

Z-N Second Method

- Step 3: With the critical values K_{cr} and P_{cr} validate the sustainable oscillation, one can obtain the PID controller according the Table given below

Table 2: Ziegler – Nichols Tuning Rule Based on Critical Values K_{cr} and P_{cr} (Second Method).

Type of Controller	K_p	T_i	T_d
P	$0.5K_{cr}$	∞	0
PI	$0.45K_{cr}$	$\frac{P_{cr}}{1.2}$	0
PID	$0.6K_{cr}$	$0.5P_{cr}$	$0.125P_{cr}$

With two same zeros and one Integral, one can find the controller that

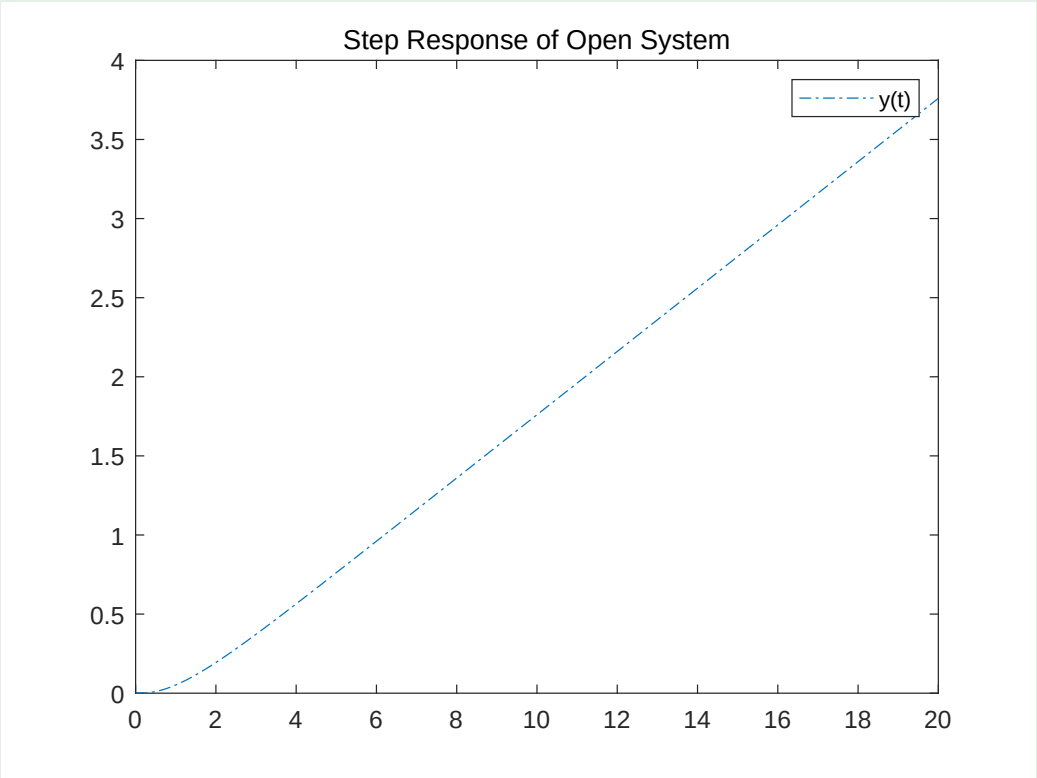
$$\begin{aligned} G_c(s) &= K_p\left(1+\frac{1}{T_i s}+T_d s\right) \\ &= 0.6K_{cr}\left(1+\frac{1}{0.5P_{cr}s}+0.125P_{cr}s\right) \\ &= 0.075K_{cr}P_{cr}\frac{\left(s+\frac{4}{P_{cr}}\right)^2}{s} \end{aligned} \tag{5}$$

Example

Consider a three orders plant $G_p(s)$ and the controller $G_C(s)$

$$G_p = \frac{1}{s(s+1)(s+5)}, G_c = K_p\left(1 + \frac{1}{T_i s} + T_d s\right)$$

Step1: The system is with an integral whose unit step response is follow and not with 'S' shape.



Example

Step2: The responses under negative feedback gain $K_p = 30$, whose oscillation period $P_{cr} \approx 2.8$

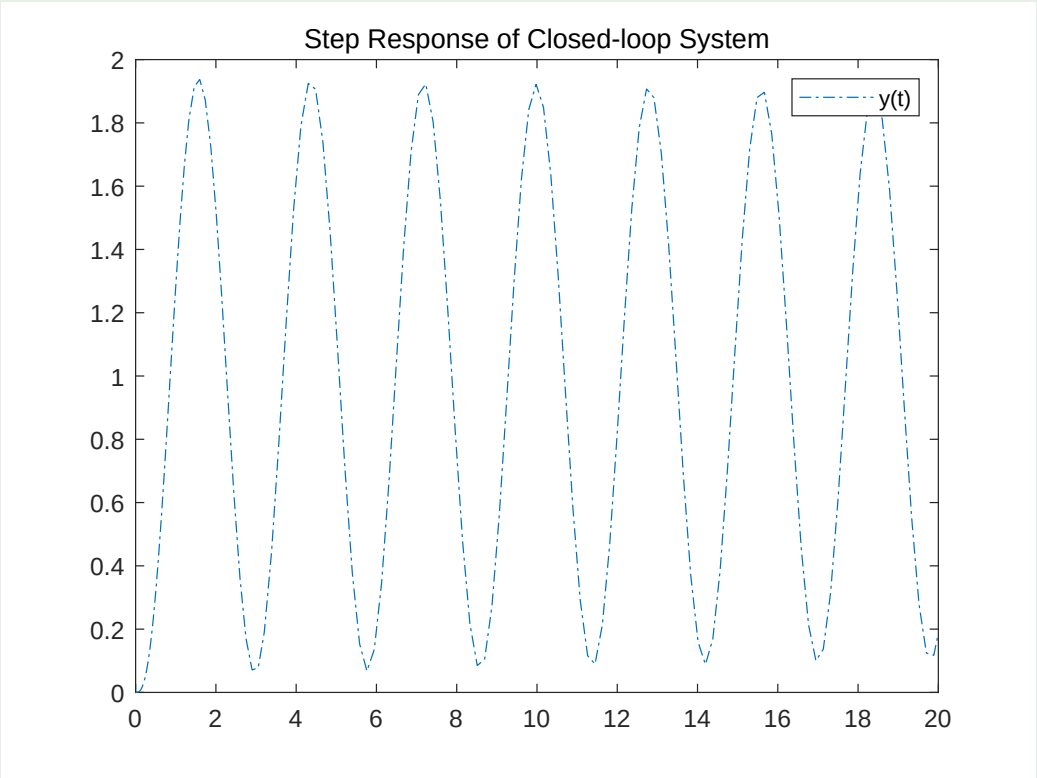


Figure 1: Oscillation of Closed-loop System

北京理工大学
BEIJING INSTITUTE OF TECHNOLOGY

Example

Step3: According to Table.2, the PID controller can be designed in P, PI, and PID forms.

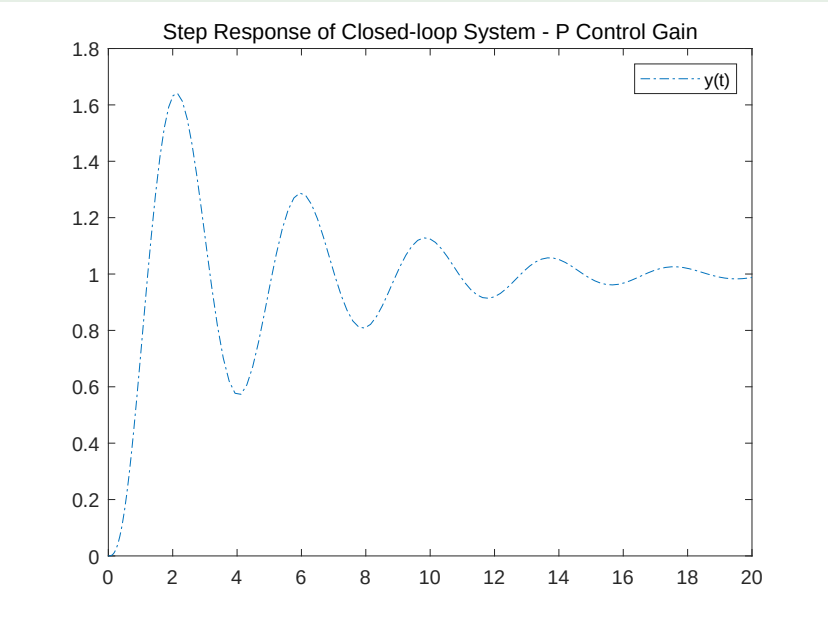


Figure 2: Step Response under P Controller

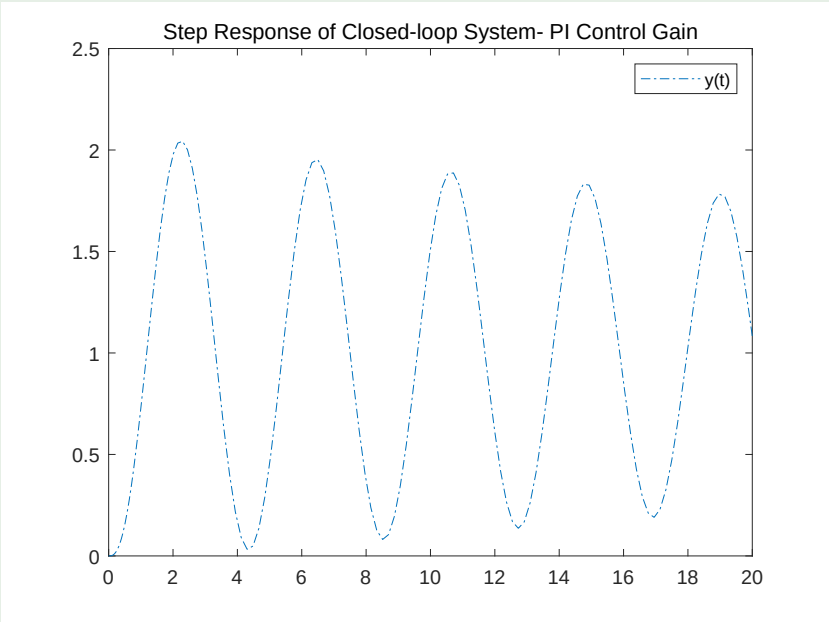


Figure 3: Step Response under PI Controller

Example

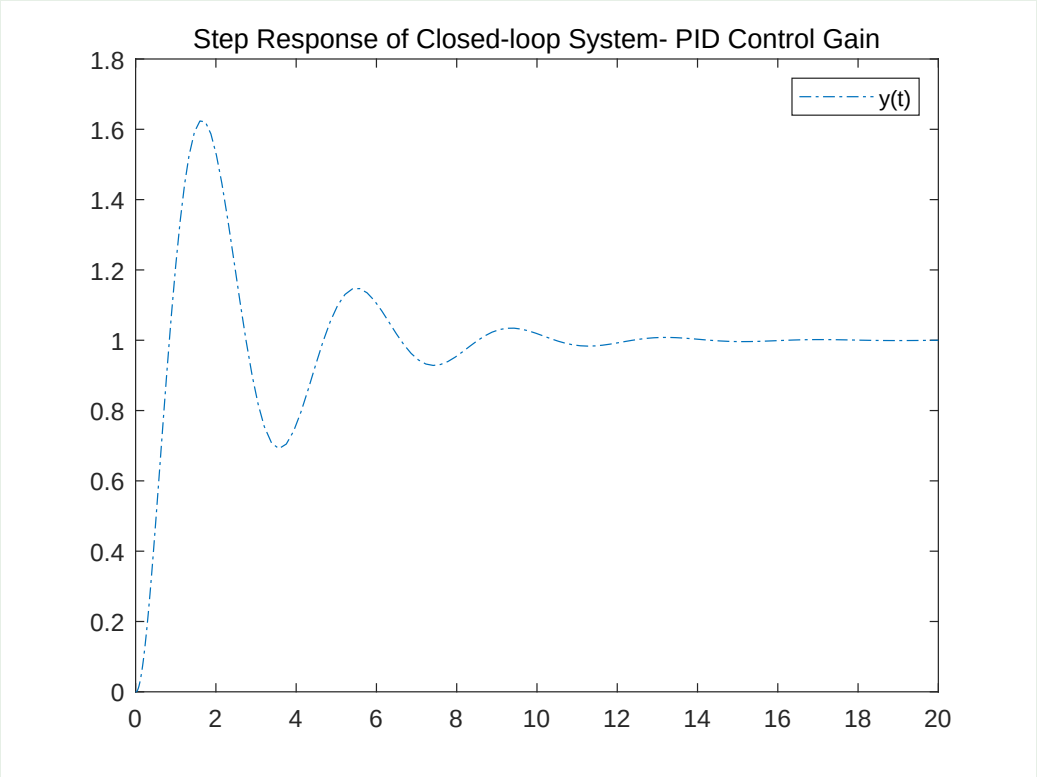
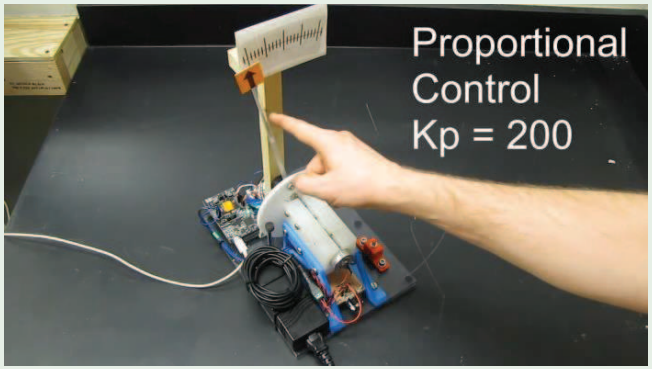


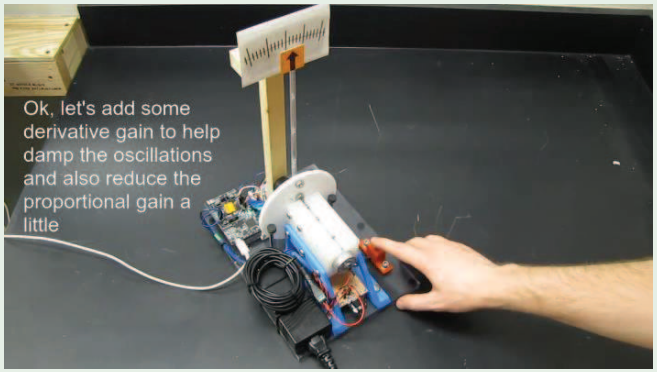
Figure 4: Step Response of Closed-loop System with PID Controller

Example

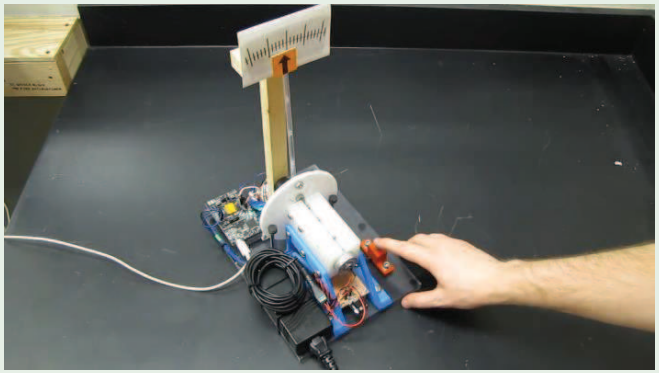
One experiment for a single degree of a pendulum with electrical motor. There is only one sensor fixed on the motor, an angular position sensor. Its system model is unknown. The control aim is to force the pendulum upward.



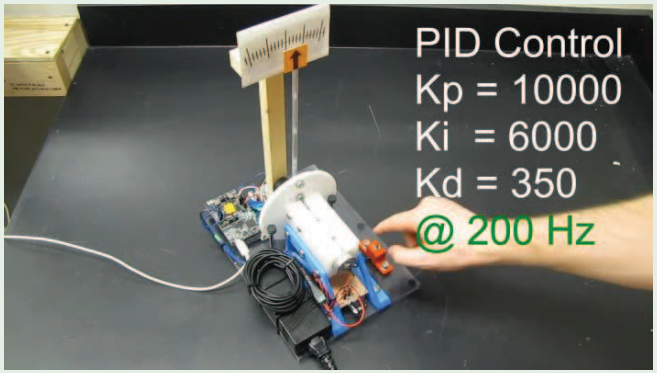
Step1： Kp for oscillations.



Step2： PD



Step3： PID+Antiwindup(Trick)



Step4： Sampling Period

Example

- Step 1. Set the proportional gain from 200，to 500. then to 1000，it begins to oscillate. So the k_p value is obtained approximately. While, if your control aim is to keep the pendulum in this state, the controller tuning process terminates.
- Step 2. as in the 2_{nd} column in the table of second method, the researcher begins to adjust the gain of the D part.
- Step 3. The PID controller can be obtained
- Step 4. A comparison of different sampling rates and the same control gain.

Example

- Step 1. Set the proportional gain from 200，to 500. then to 1000，it begins to oscillate. So the k_p value is obtained approximately. While, if your control aim is to keep the pendulum in this state, the controller tuning process terminates.
- Step 2. as in the 2_{nd} column in the table of second method, the researcher begins to adjust the gain of the D part.
- Step 3. The PID controller can be obtained
- Step 4. A comparison of different sampling rates and the same control gain.

When the sampling and control frequency is 200, the motor is with strong noise. The pendulum vibrates heavily.
Setting the sampling frequency as 800, the noise is much weak. The pendulum doesn't vibrate.



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

Example

- Step 1. Set the proportional gain from 200，to 500. then to 1000，it begins to oscillate. So the k_p value is obtained approximately. While, if your control aim is to keep the pendulum in this state, the controller tuning process terminates.
- Step 2. as in the 2_{nd} column in the table of second method, the researcher begins to adjust the gain of the D part.
- Step 3. The PID controller can be obtained
- Step 4. A comparison of different sampling rates and the same control gain.

When the sampling and control frequency is 200, the motor is with strong noise. The pendulum vibrates heavily.
Setting the sampling frequency as 800, the noise is much weak. The pendulum doesn't vibrate.

In this experiment, the researcher doesn't calculate the critical gain and period accurately. The estimated critical values are employed in the controller, and some tiny improvements are made in the tuning process. Maybe the controller is not the best one, but a useful one.



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

1 Introduction

2 Ziegler - Nichols Rules

3 Design of PID Controllers with Frequency - Response Approach

4 PID Controllers with Computational Optimization Approach

5 Modification of PID Control Schemes

6 Conclusion

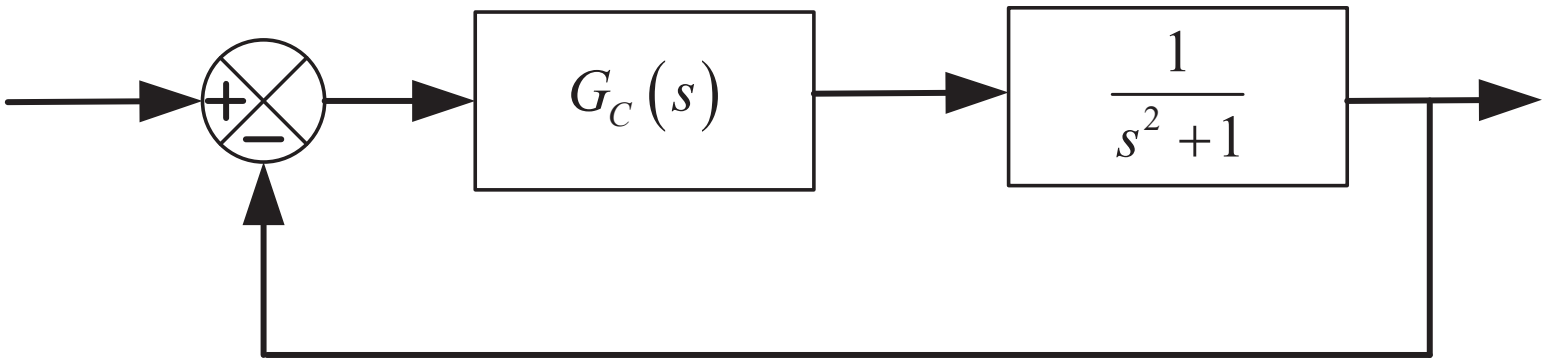


Figure 5: Plant under Controller $G(s)$

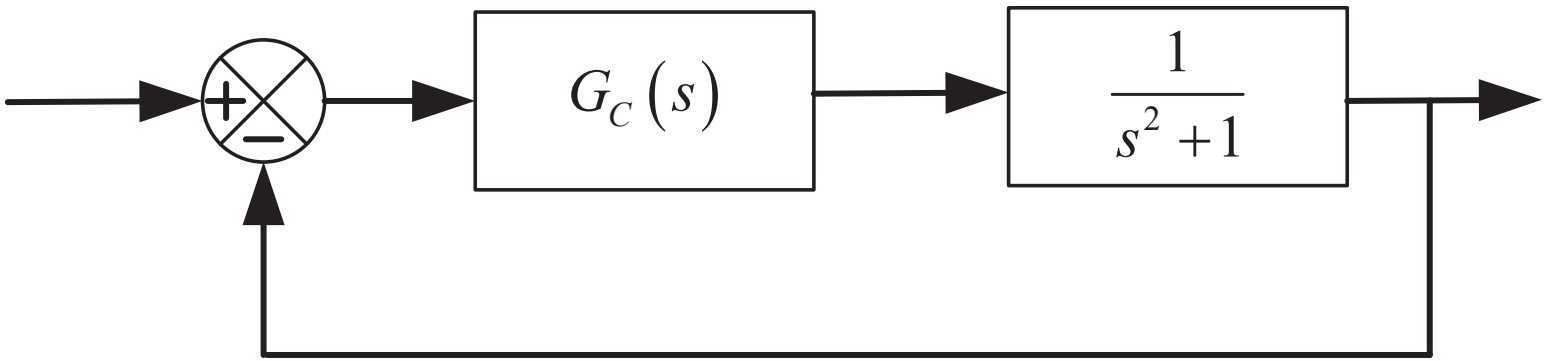


Figure 5: Plant under Controller $G(s)$

Consider a plant in Fig.5, the target here is to design a PID controller $G_c(s)$ that

Control Target

1. Static error constant is 4
2. Phase margin $\geq 50°$
3. Gain margin ≥ 10 **dB**

Choose the PID controller in the below form

$$G_c(s) = \frac{K(as+1)(bs+1)}{s} \tag{6}$$

The design steps can be described as following

Frequency based Design Steps

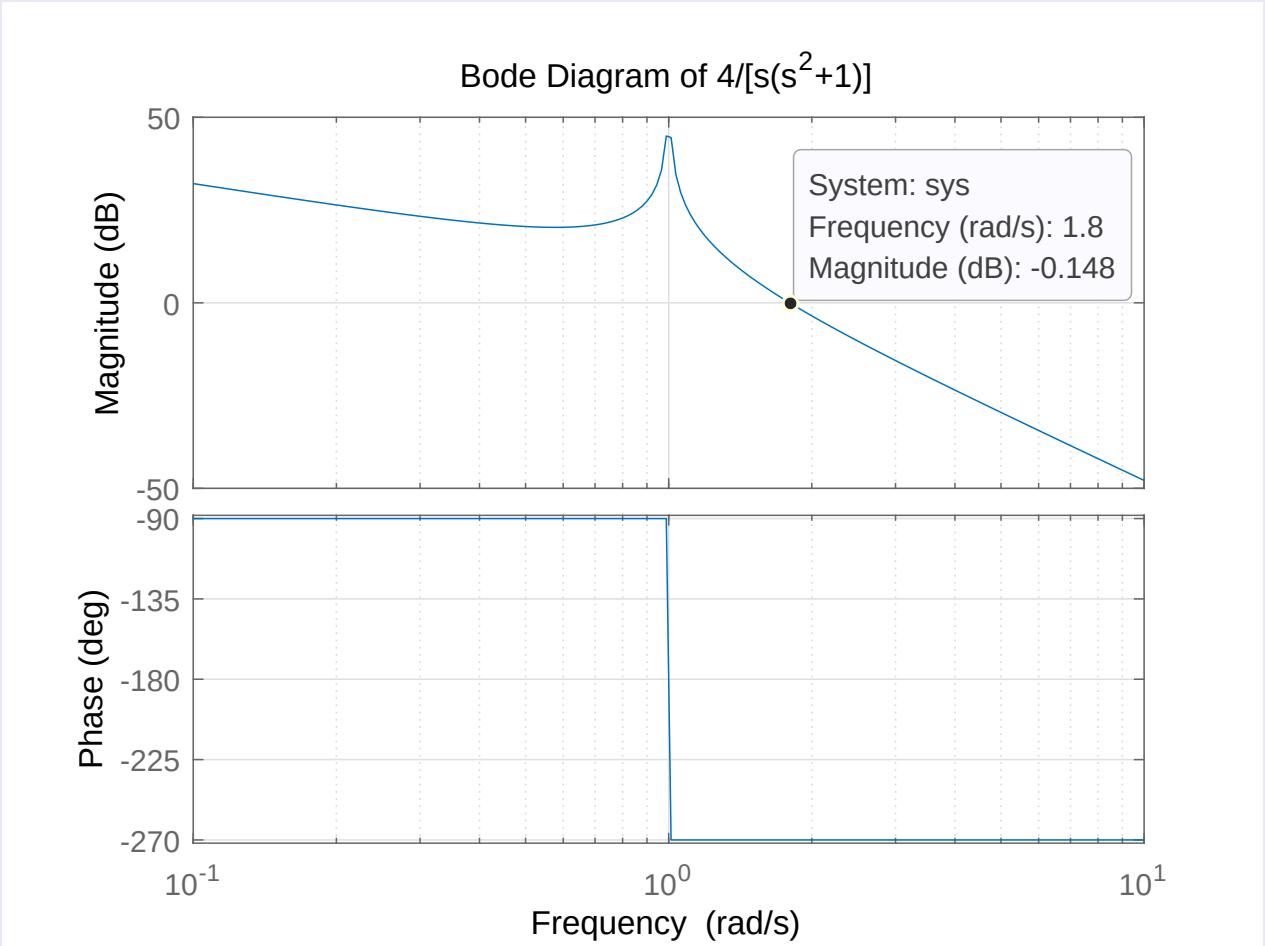
- Step 1: Since the static error constant is specified as 4, it has

$$\begin{aligned} K &= \lim_{s\to 0} sG_c(s)\frac{1}{s^2+1} \\ &= \lim_{s\to 0} s\frac{K(as+1)(bs+1)}{s}\frac{1}{s^2+1} \\ &= 4 \end{aligned} \tag{7}$$

Frequency based Design Steps

- Step 2: Plot bode diagram

$$G(s) = \frac{4}{s(s^2+1)} \tag{8}$$



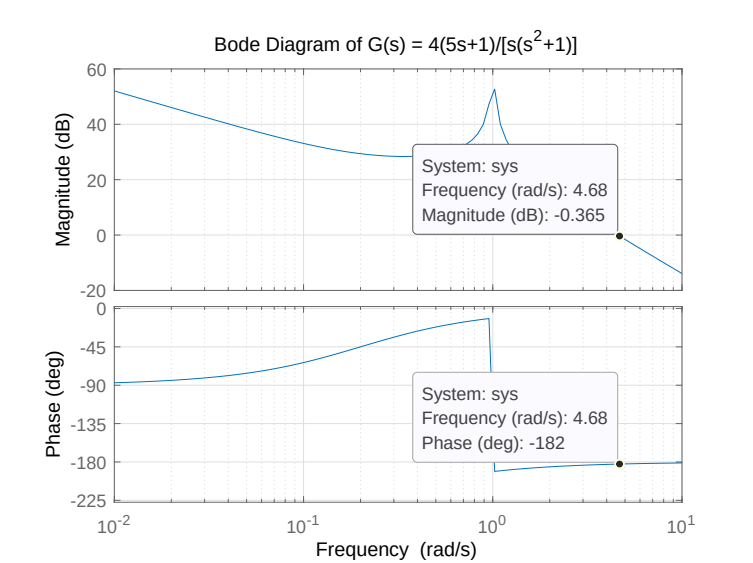
Frequency based Design Steps

- Step 3: Consider the phase margin $\geq 50^\circ$ and gain margin $\geq 50dB$. It finds the gain crossover frequency is approximately $\omega = 1.8rad \cdot sec^{-1}$. Assume the gain crossover frequency of the compensated system to be in $\left[1rad \cdot sec^{-1}, 10rad \cdot sec^{-1}\right]$, it has

$$G_c(s) = \frac{4(as+1)(bs+1)}{s} \tag{9}$$

One chooses $a = 5$. $as + 1$ contributes up to 90° phase lead in the high frequency region. The Bode diagram of

$$G(s) = \frac{4(5s+1)}{s(s^2+1)} \tag{10}$$



Frequency based Design Steps

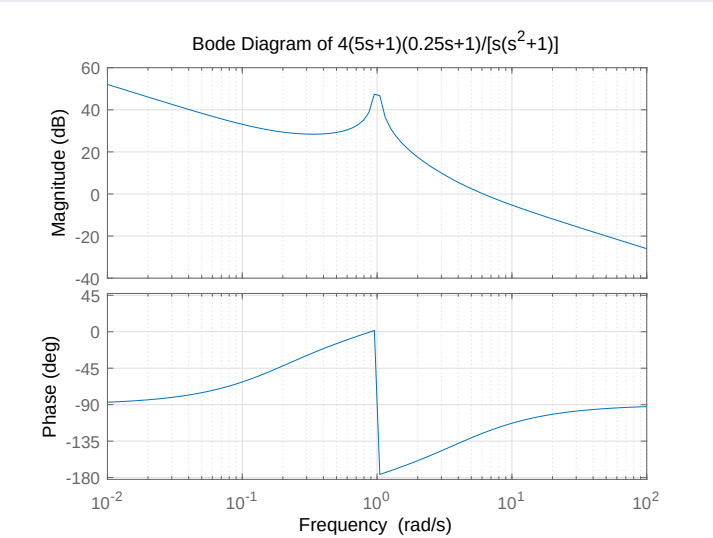
- Step 4: Reconsider the target of phase margin $\geq 50^\circ$ and gain margin $\geq 50dB$. The parameter b of term $bs+1$ is to give. One chooses $b = 0.25$, it has

$$G_c(s) = \frac{4(5s+1)(0.25s+1)}{s} \tag{11}$$

the open-loop transfer function is

$$G(s) = \frac{4(5s+1)(0.25s+1)}{s(s^2+1)} \tag{12}$$

It has the bode plot below whose phase margin is about 60°, and gain margin is $+\infty$.



Frequency based Design Steps

- Step 5: The closed-loop transfer function is

$$\frac{G(s)}{1+G(s)} = \frac{5s^2+21s+4}{s^3+5s^2+22s+4} \tag{13}$$

The unit-step and ramp-step responses are in Fig.6 and Fig.7

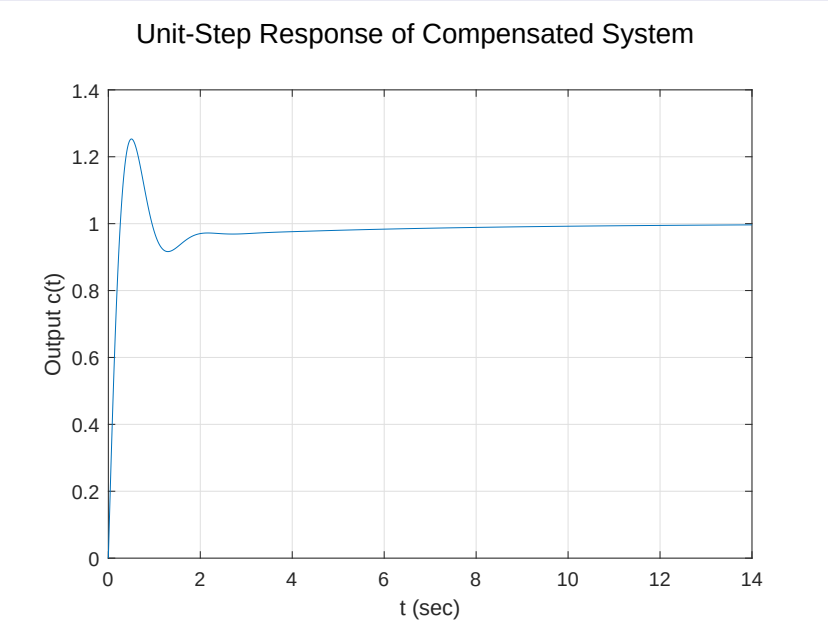


Figure 6: Step Response of Closed-loop System

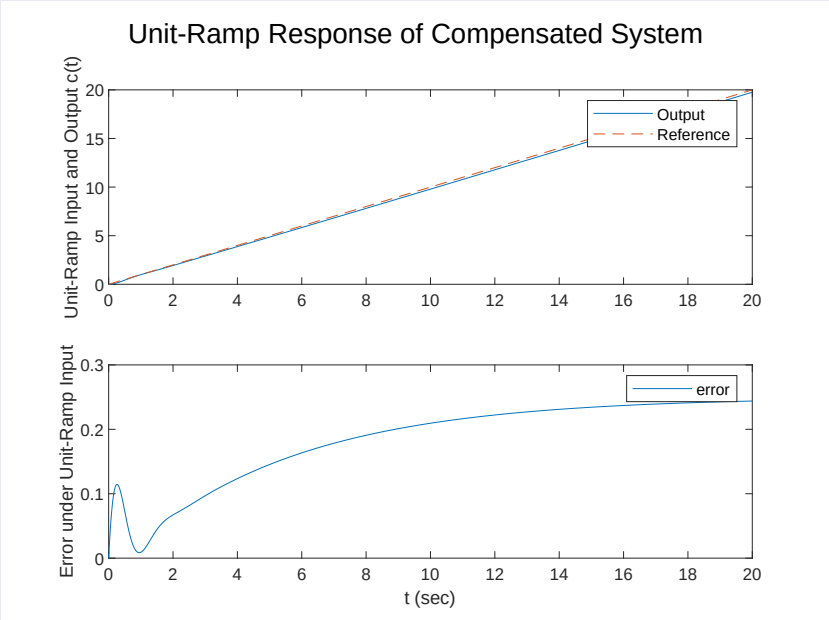


Figure 7: Ramp Response of Closed-loop System

1 Introduction

2 Ziegler - Nichols Rules

3 Design of PID Controllers with Frequency - Response Approach

4 PID Controllers with Computational Optimization Approach

5 Modification of PID Control Schemes

6 Conclusion



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

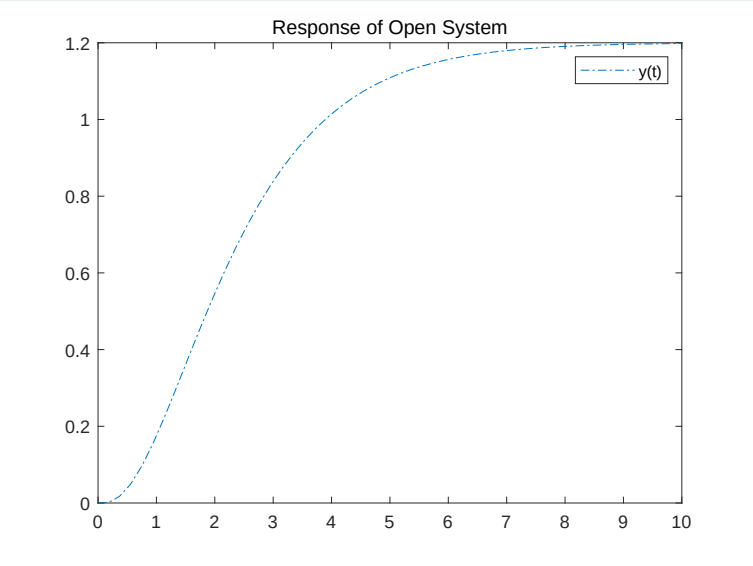
Optimal gain selection according to the performance of the closed-loop system.

Example

Consider a known system below

$$G(s) = \frac{1.2}{0.36s^3 + 1.86s^2 + 2.5s + 1} \tag{14}$$

whose output performance is



Example

The control target is: maximum overshoot in the unit-step response $\leq 10\%$ under the controller below

$$G_c(s) = K\frac{(s+a)^2}{s} \tag{15}$$

There exist not only solution satisfying the requirement. Assume the search region of K and a to be

$$K \in [2,3]\,, a \in [0.5, 1.5] \tag{16}$$

in the computational approach, the step searching to be ΔK and Δa. Set that

$$\begin{aligned} K_i &= \Delta K(i-1) + \min K \\ a_j &= \min a + \Delta a(j-1) \end{aligned}$$

where

$$\begin{aligned} i &= 1, 2, \cdots, N_K, N_K = \frac{\max K - \min K}{\Delta K} + 1 \\ j &= 1, 2, \cdots, N_a, N_a = \frac{\max a - \min a}{\Delta a} + 1 \end{aligned}$$

and see the step response under varied K_i and a_j.

Example

The systems whose overshoot range from 1 to 1.1 are selected and listed in Fig.8

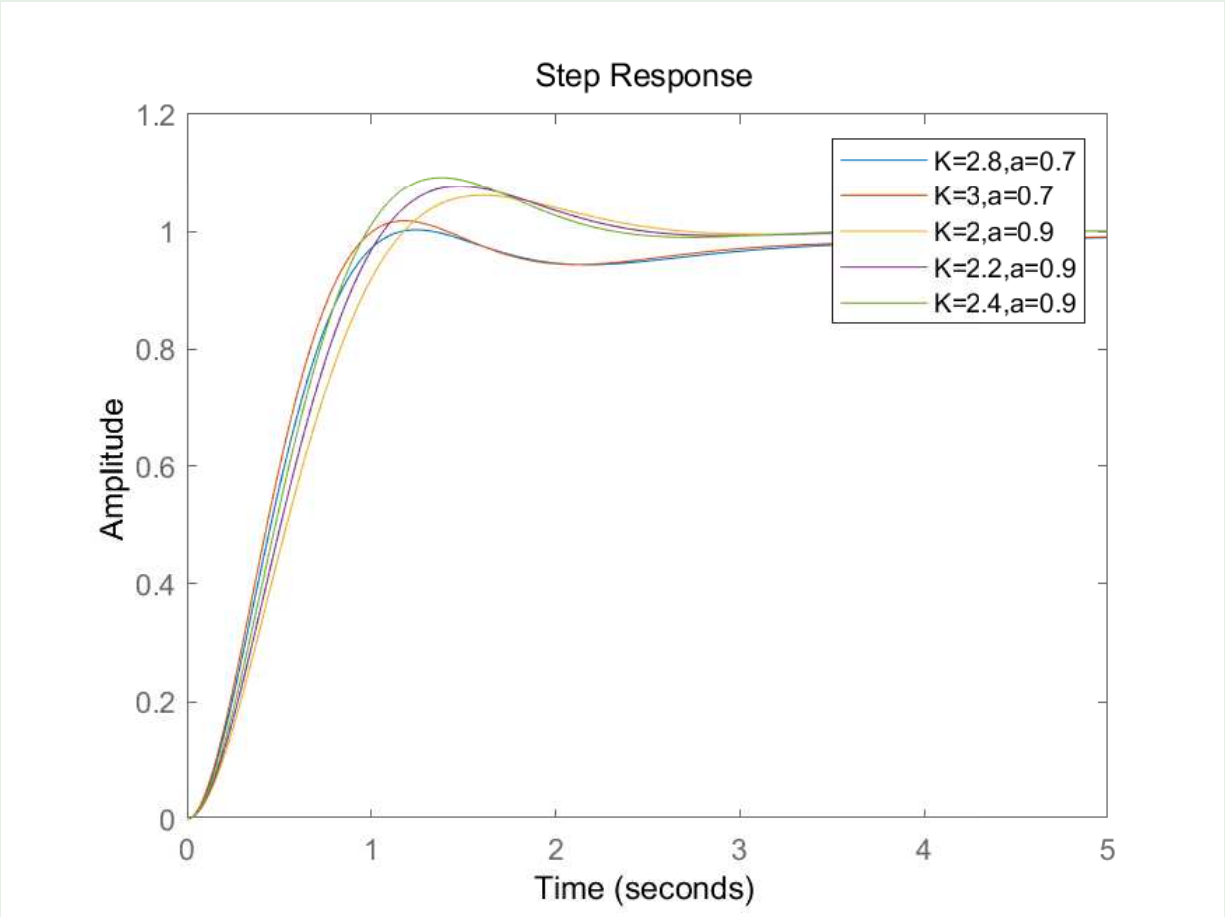


Figure 8: Responses of Closed-loop System

1 Introduction

2 Ziegler - Nichols Rules

3 Design of PID Controllers with Frequency - Response Approach

4 PID Controllers with Computational Optimization Approach

5 Modification of PID Control Schemes
- Modification on Derivative
- Modification on PID
  - PI-D Controller
  - I-PD Controller
  - LPF based Reference Input

6 Conclusion



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

It's unexpected to employ the derivative block directly.

It's unexpected to employ the derivative block directly.

Why?

It's unexpected to employ the derivative block directly.

Why?

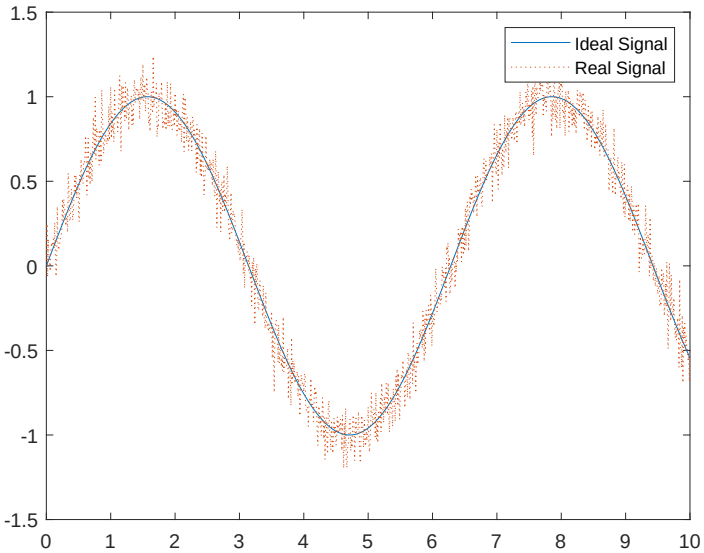


Figure 9: Ideal Signal and Real Signal

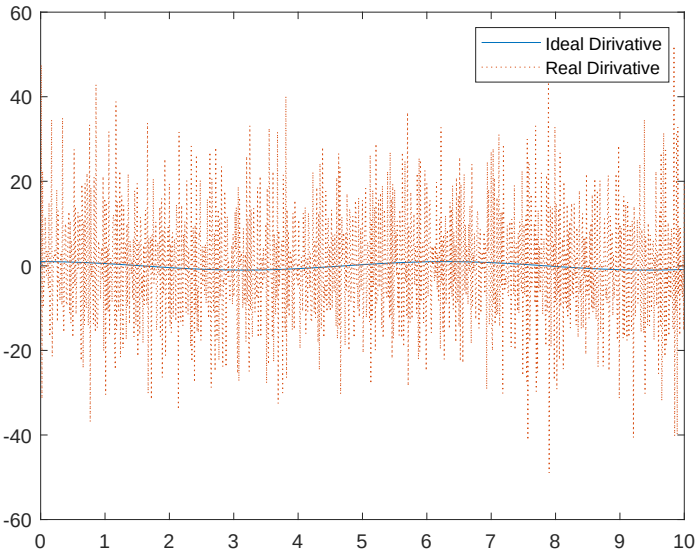


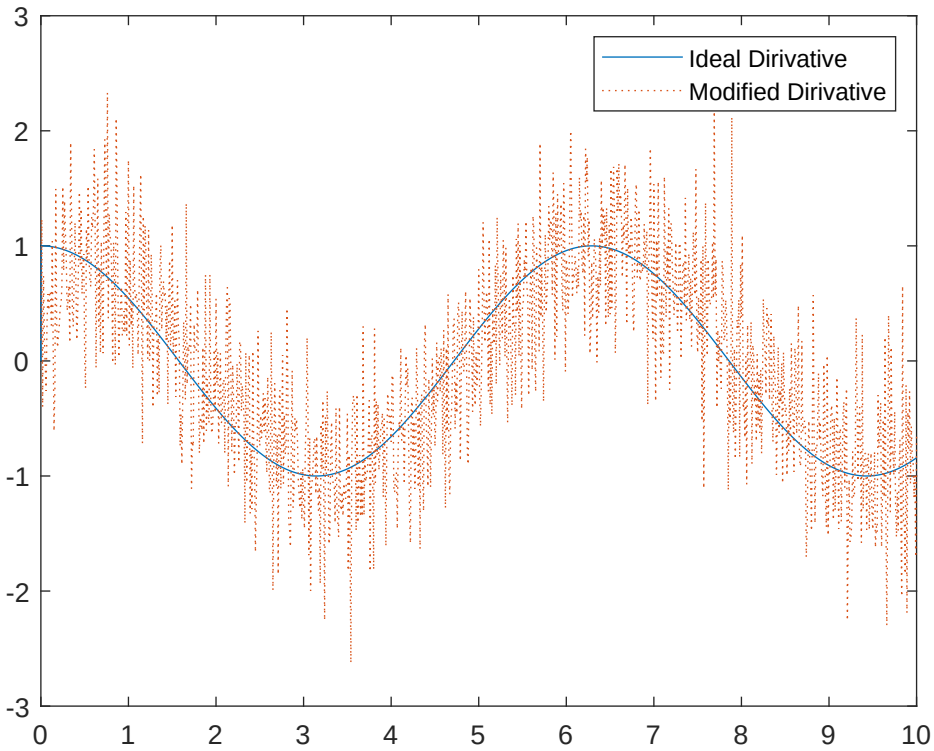
Figure 10: Ideal and Real Derivative

What We can Do?

Instead of using derivative block directly, the modified derivative in equation (17), is introduced.

$$\frac{s}{\tau_d s + 1} \tag{17}$$

where τ_d is normally selected as $\tau_d \ll 1$.



Homework

Try to obtain a more smoothed signal via changing τ_d and connecting (17) serially.

北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

Example

Consider a system plant below

$$G(s) = \frac{1}{s(s+1)(s+5)} \tag{18}$$

the reference input is assumed to be a unit-step signal. The PID controller is in the following form

$$G_c(s) = K_p + K_i \frac{1}{s} + K_d s$$

whose parameters are $K_p = 18$ $K_i = 12.811$, and $K_d = 6.322$. The modified derivative is designed as

$$\frac{s}{0.2s+1}$$



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

Example

As in Fig.11, the output result under the ideal controller is shown. It can be seen that the system output approaches zero in about ten seconds.

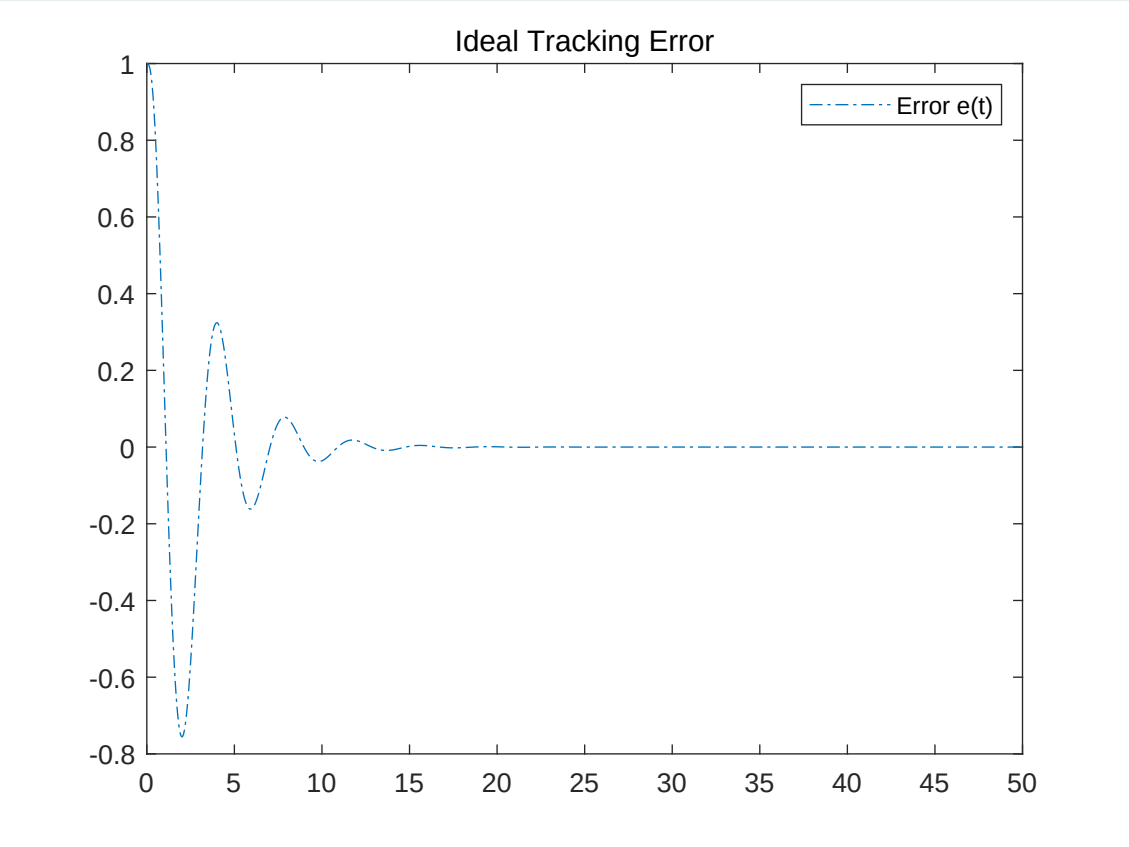


Figure 11: The Ideal Tracking Error

1 Introduction

2 Ziegler - Nichols Rules

3 Design of PID Controllers with Frequency - Response Approach

4 PID Controllers with Computational Optimization Approach

5 Modification of PID Control Schemes
- Modification on Derivative
- Modification on PID
  - PI-D Controller
  - I-PD Controller
  - LPF based Reference Input

6 Conclusion

北京理工大學 BEIJING INSTITUTE OF TECHNOLOGY

Consider a system plant below

$$G(s) = \frac{50s + 50}{s^2 + 5s + 6} \tag{19}$$

whose controller is in PID form that

$$G_c(s) = 1.3132 + 0.1637\frac{1}{s} + 0.0046s \tag{20}$$

the reference signal is assumed to be

$$r(t) = \begin{cases} 0, \text{ if } t < 1 \\ 1, \text{ if } t >= 2 \end{cases}$$



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

Consider a system plant below

$$G(s) = \frac{50s + 50}{s^2 + 5s + 6} \tag{19}$$

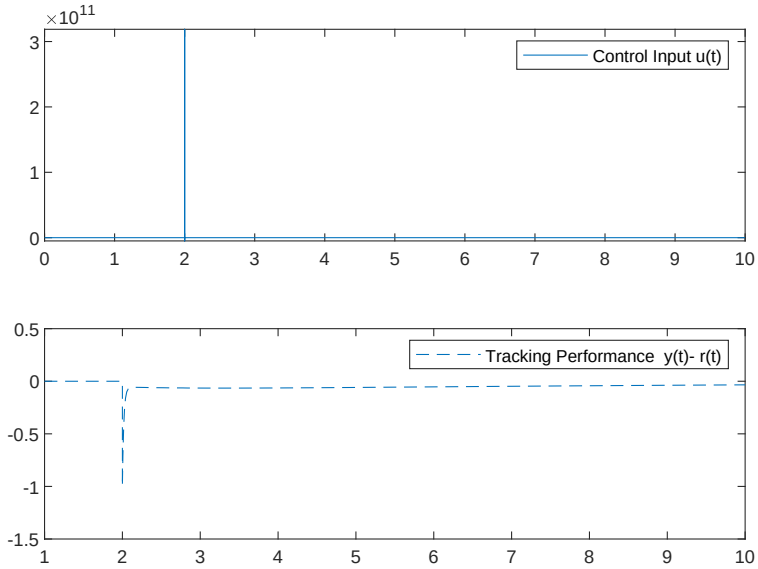
whose controller is in PID form that

$$G_c(s) = 1.3132 + 0.1637\frac{1}{s} + 0.0046s \tag{20}$$

the reference signal is assumed to be

$$r(t) = \begin{cases} 0, & \text{if } t < 1 \\ 1, & \text{if } t >= 2 \end{cases}$$

Impulsive Control and The Tracking Performance



Consider a system plant below

$$G(s) = \frac{50s + 50}{s^2 + 5s + 6} \tag{19}$$

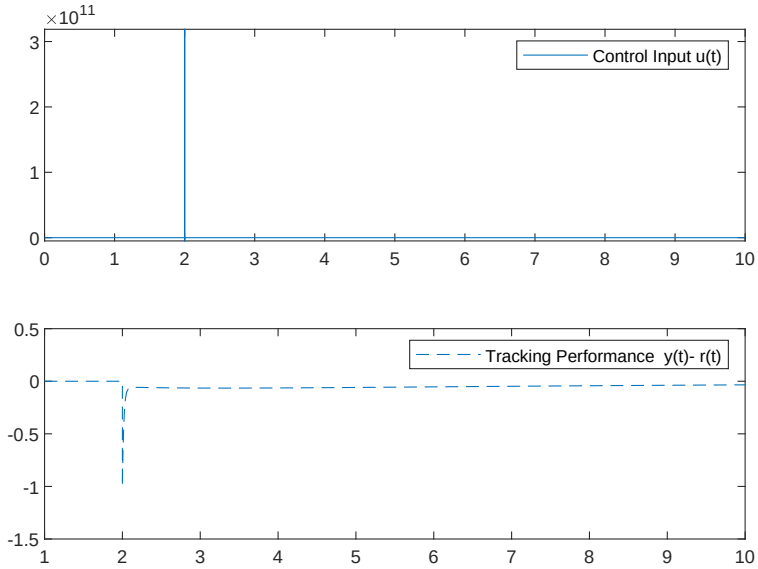
whose controller is in PID form that

$$G_c(s) = 1.3132 + 0.1637\frac{1}{s} + 0.0046s \tag{20}$$

the reference signal is assumed to be

$$r(t) = \begin{cases} 0, & \text{if } t < 1 \\ 1, & \text{if } t >= 2 \end{cases}$$

Impulsive Control and The Tracking Performance



Problems

- High amplitude impulsive signal for input, danger.
- Saturate the actuator.

Solutions: modified PI-D, I-PD, and LPF reference.

北京理工大學 BEIJING INSTITUTE OF TECHNOLOGY

5 Modification of PID Control Schemes

- Modification on Derivative
- Modification on PID
  - PI-D Controller
  - I-PD Controller
  - LPF based Reference Input



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

PI-D

$$U(s) = (R(s) - Y_N(s)) \left(K_p + K_i \frac{1}{s} \right) - \left(K_p + K_i \frac{1}{s} + K_d s \right) Y_N(s) \tag{21}$$

where $U(s)$ is the input Laplace function, $Y_N(s)$ is the output signal mixed with noise, and $R(s)$ is the reference signal, the parameters of PID controller are the same as in $G_C(s)$.

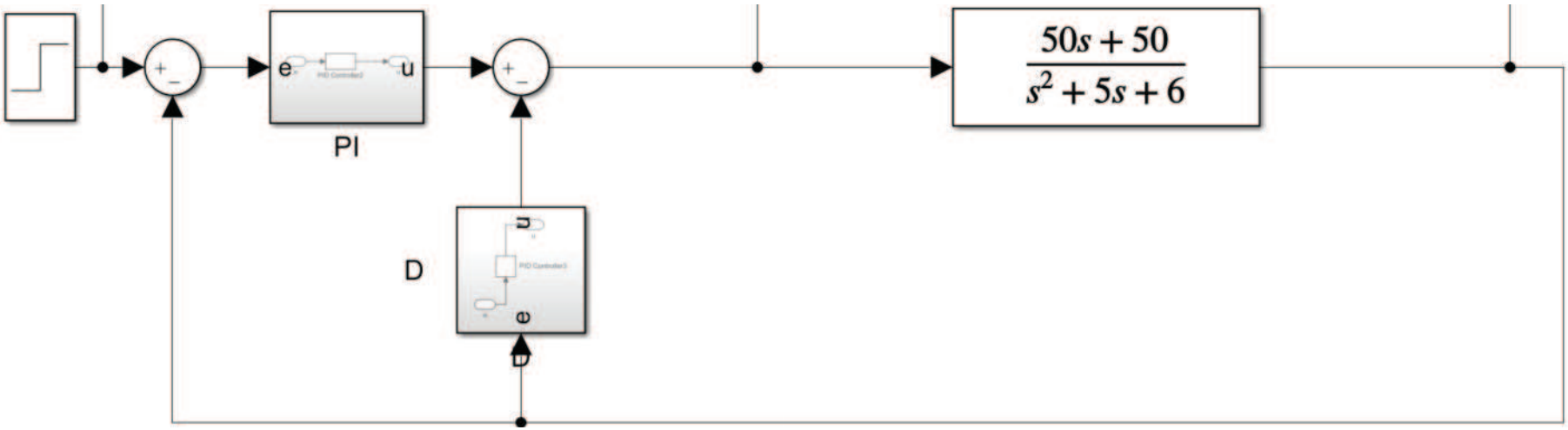


Figure 12: PI - D Structure

The control input still contains impulse, but much lower amplitude, the output tracks the reference signal well.

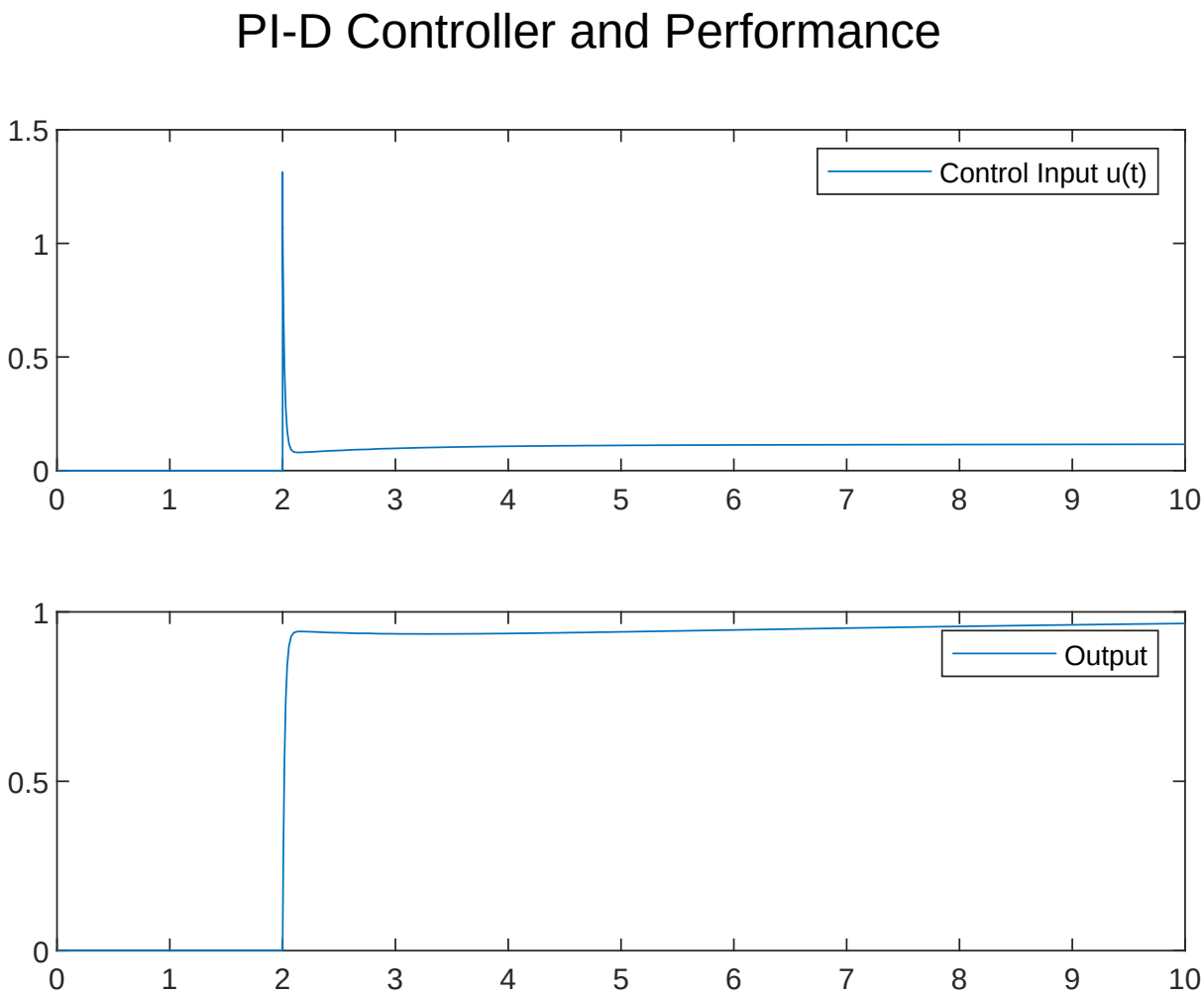


Figure 13: PI - D Controller and Performance

5 Modification of PID Control Schemes

- Modification on Derivative
- Modification on PID
  - PI-D Controller
  - I-PD Controller
  - LPF based Reference Input



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

I-PD controller

$$U(s) = (R(s) - Y_N(s)) \left(K_p \frac{1}{s} \right) - \left(K_i \frac{1}{s} + K_d s \right) Y_N(s) \tag{22}$$

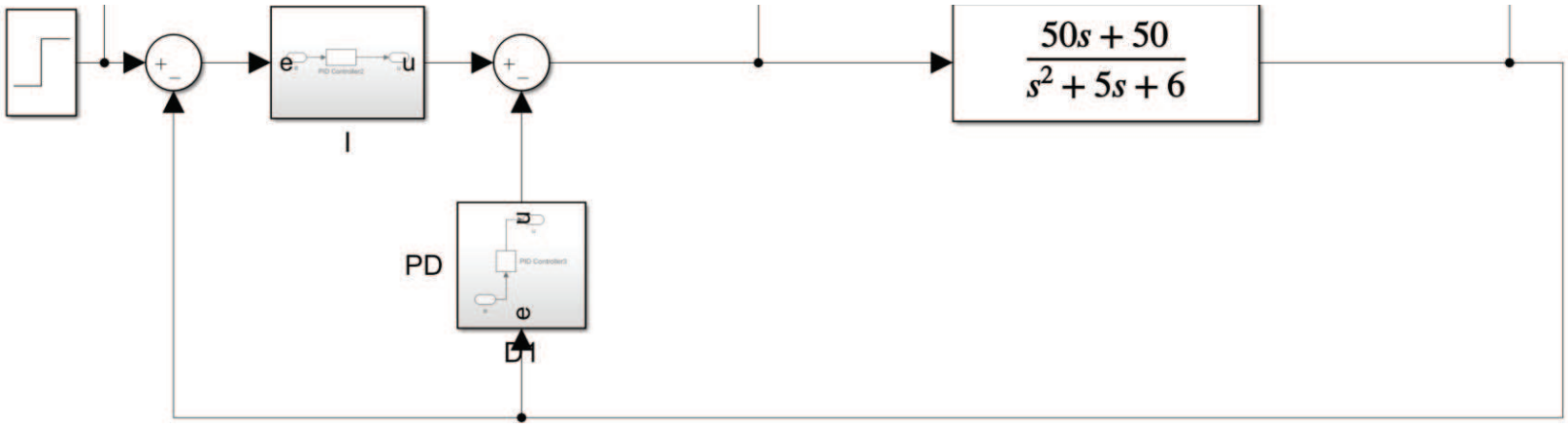


Figure 14: I - PD

The control input to the plant system of I-PD and the output performance are depicted in Fig.15. The impulse can be eliminated completely, and the output tracks the reference signal very slowly. Not proper to be used directly.

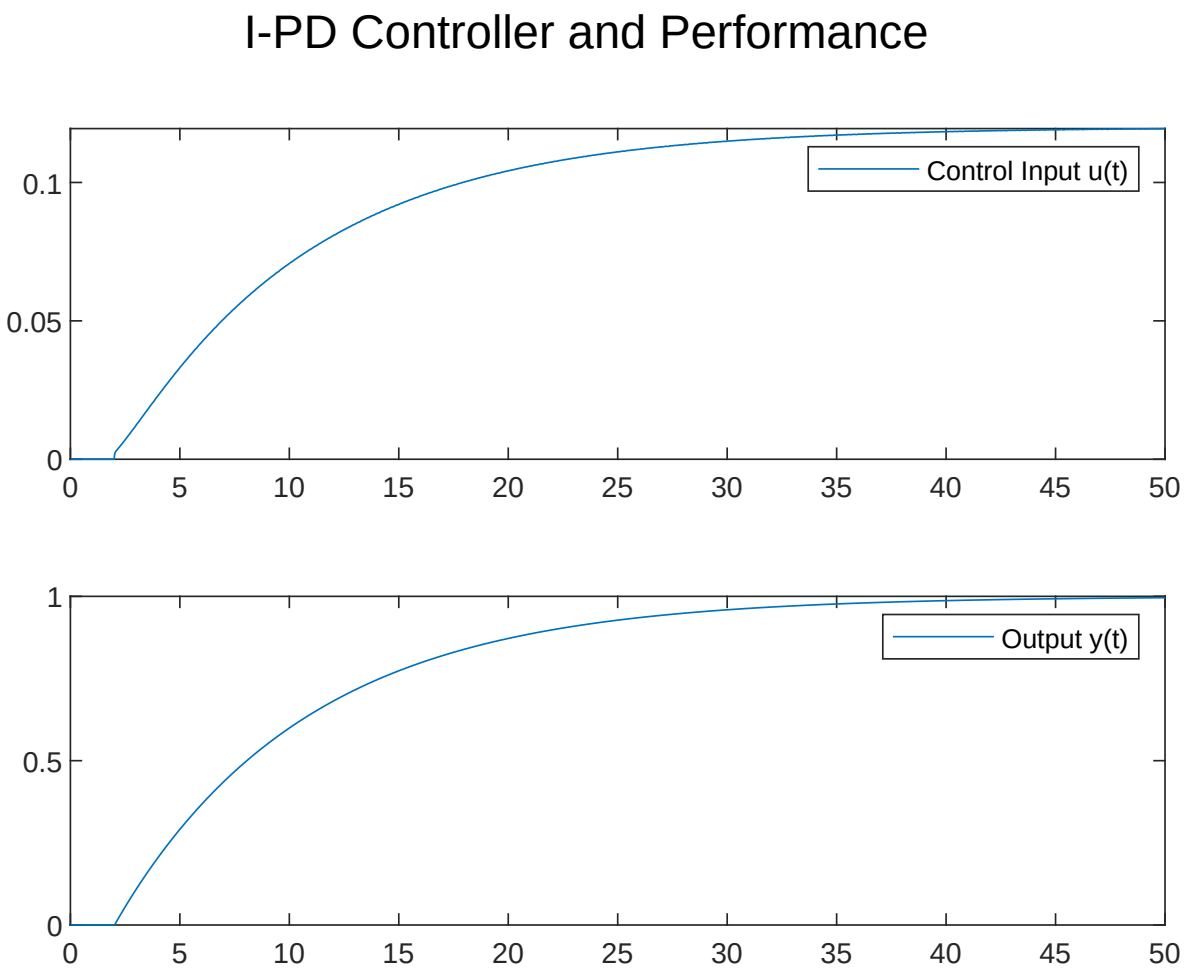


Figure 15: I - PD Controller and Performance

5 Modification of PID Control Schemes

- Modification on Derivative
- Modification on PID
  - PI-D Controller
  - I-PD Controller
  - LPF based Reference Input



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

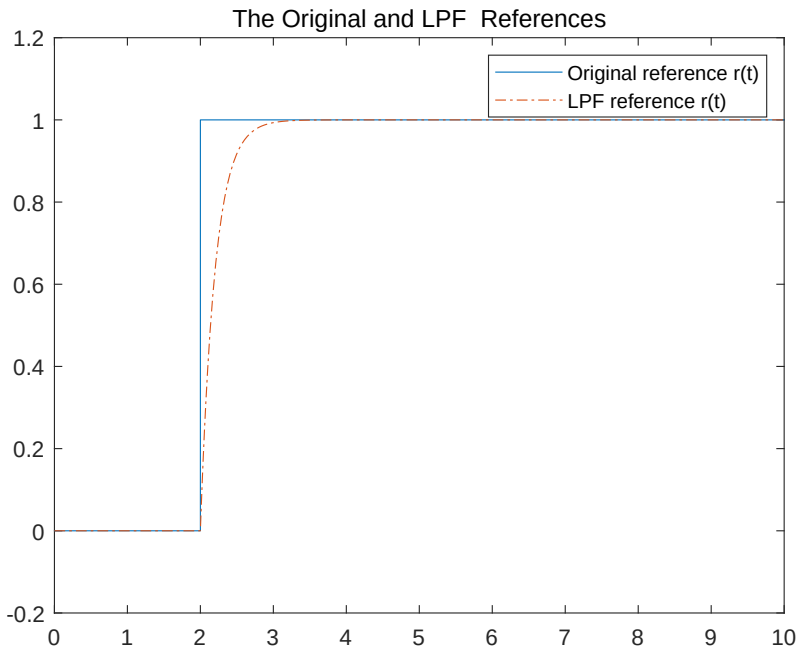


Figure 16: LPF based Reference Control Structure



Figure 16: LPF based Reference Control Structure

The reference applied to the input node is not so sharp.



The control input and output performance are depicted as in Fig.18, the output signal is similar as in the PI-D, but the control input to the plant is not with impulse anymore.

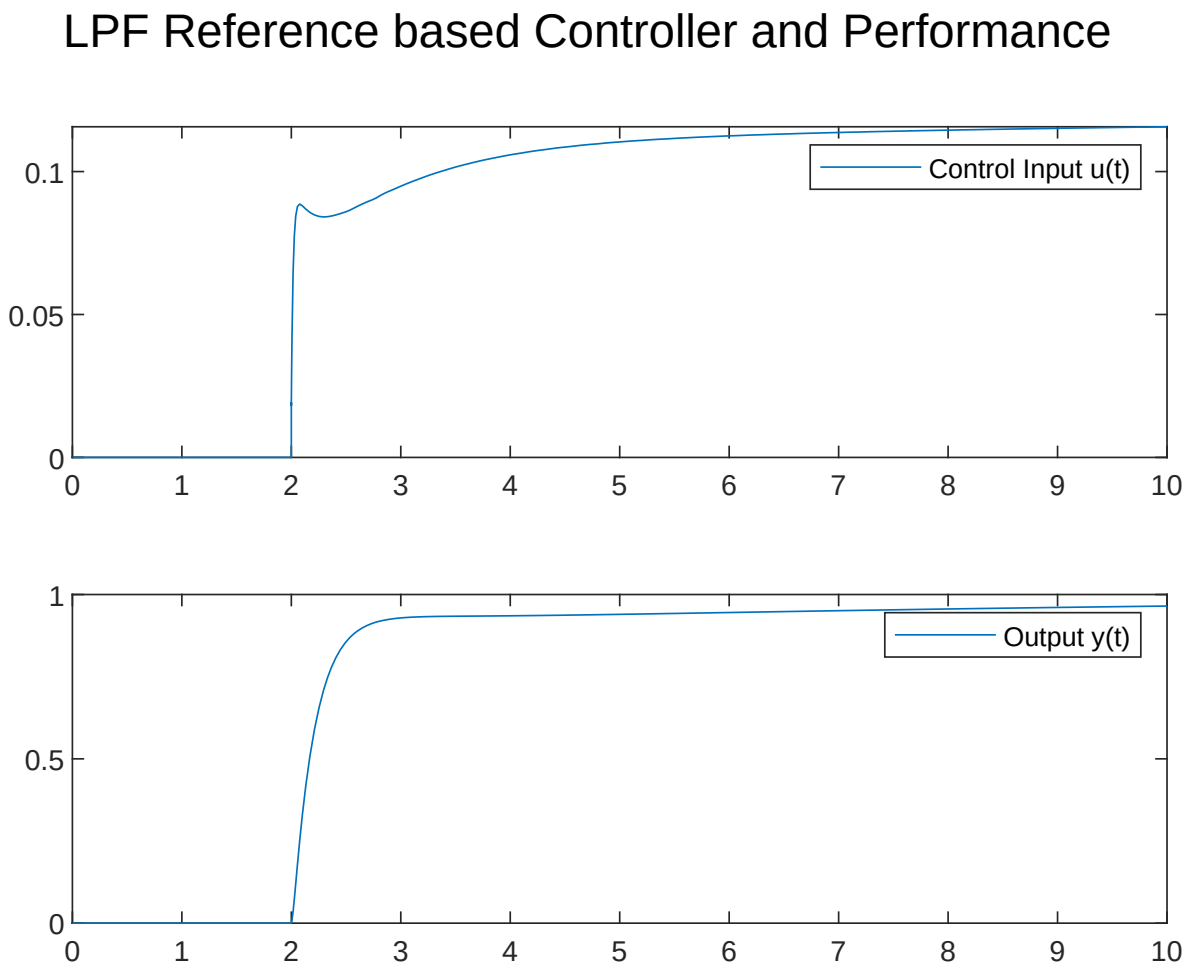


Figure 18: LPF Reference based Control and Output Performance

1 Introduction

2 Ziegler - Nichols Rules

3 Design of PID Controllers with Frequency - Response Approach

4 PID Controllers with Computational Optimization Approach

5 Modification of PID Control Schemes

6 Conclusion



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

PID Controller

- Ziegler – Nichols Rules
  - 1st Z-N Rule: S shape
  - 2nd Z-N Rule: Can make response oscillation
- Frequency – Constant Error, Phase Margin and Gain Margin;
- Computational Optimization



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

PID Controller

- Ziegler – Nichols Rules
  - 1st Z-N Rule: S shape
  - 2nd Z-N Rule: Can make response oscillation
- Frequency – Constant Error, Phase Margin and Gain Margin;
- Computational Optimization

Don't Employ the Derivative Block Directly!



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY

PID Controller

- Ziegler – Nichols Rules
  - 1st Z-N Rule: S shape
  - 2nd Z-N Rule: Can make response oscillation
- Frequency – Constant Error, Phase Margin and Gain Margin;
- Computational Optimization

Don't Employ the Derivative Block Directly!

Some Tricks

- Impulse avoid:
  - Modification PID – PI + D and I+PD
  - Low Pass Filter (LPF)
- Sampling frequency issue.

谢谢!

sunyy@bit.edu.cn



北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY